

Unit 3

Financial Independence

Unit 3: Financial Independence

What is financial independence?

How much is needed?

How can I achieve it?

What are some common strategies?

The superannuation system

Consolidating your super

Investment options

Making contributions into super

Drawing an income from super

Withdrawing funds from super

Super strategies

Discussion

Sitting by yourself? Please move next to someone else!

Say “hello” and introduce yourself in groups of 2 or 3.

One of the main topics we will be talking about is **financial independence.**

What do you think most of your friends think it means?

What do you think your parents think it means?

What do you think financial planners think it means?

General Advice Warning

The content of this Unit is not personal financial advice for you

It covers general principles

... taught as part of university course based on the Australian system.

It does not take into account your situation, objectives or needs.

You are responsible to consider whether it is suitable

... for your personal circumstances.

You should consider obtaining financial and tax advice yourself.

What is Financial Independence?

What is Financial Independence?

A situation in which the income you are drawing from your investments is enough to cover your living expenses ... indefinitely!

This means that you no longer need to do paid work

... freeing you up to do unpaid voluntary work

... or paid work that aligns with your values

... and spend a bit more time on leisure in relationship with others

It is not the same as 'being rich'

What does 'rich' actually mean?

A flashy car, house and holidays?

... these things don't generate cash flow

... and may be funded with large amounts of debt

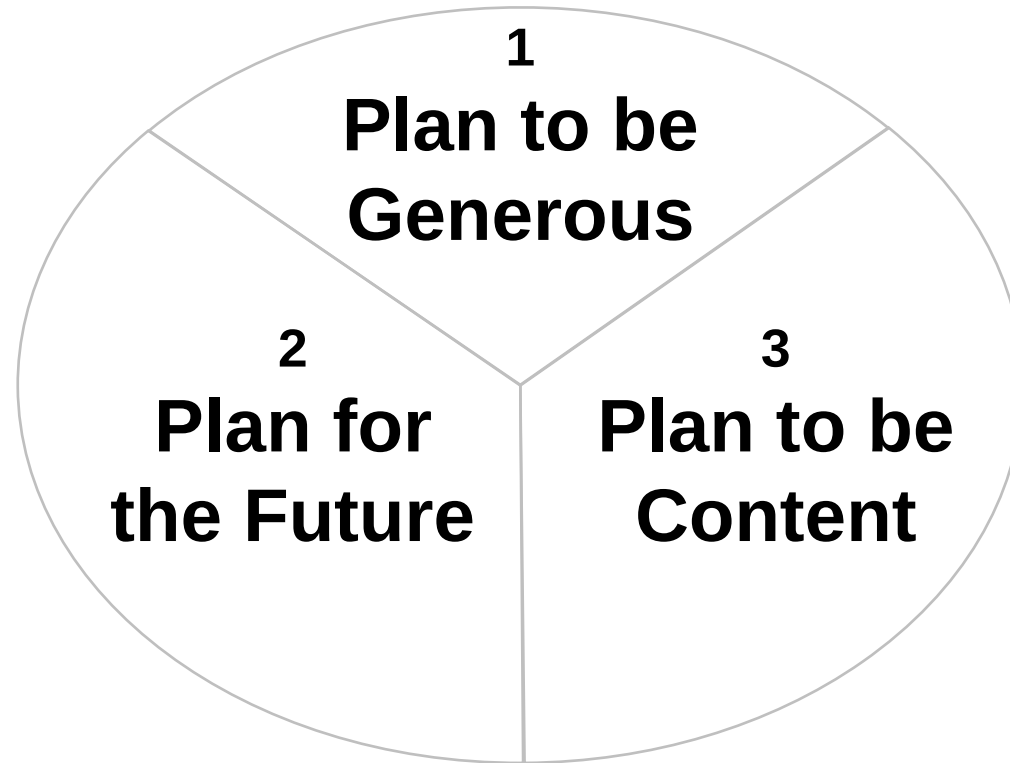
Robert Kiyosaki jokes that a car and house is really a liability

... since they are a use of cash rather than a source of cash

The Pie and Financial Independence

Your **pie** is your
total income
each month

and what you do with it!



Investment income covers all three

The Snowball is your Investment Assets

... that generate cash flow.

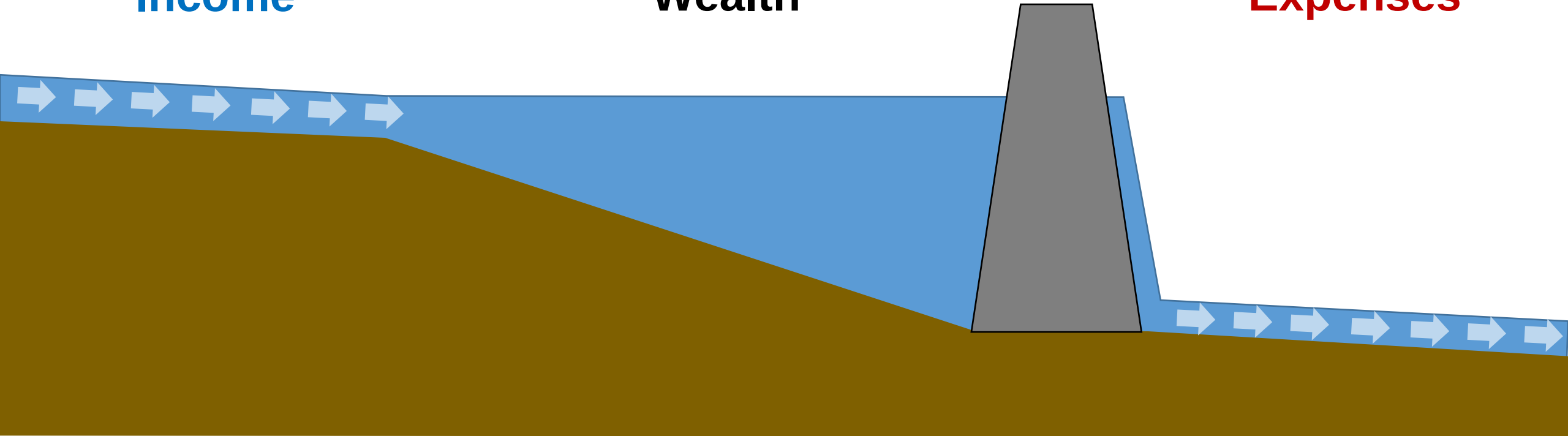
It is important to keep expenses under control so we can regularly save.

The Dam and Financial Independence

Income

Wealth

Expenses



Why not just use the term 'retirement'?

RETIREMENT

RETIREMENT

Problems with the term ‘Retirement’

- 1. Negative associations of ‘being old’ and ‘useless’**
- 2. Feels like a long way off for people below the age of 45**
- 3. ... so you don’t need to think about it or plan for it until 50+**
- 4. Sounds like you are not doing any ongoing paid work**
- 5. Sounds like you are not doing any ongoing unpaid work**

Benefits of the term ‘Financial Independence’

- 1. Positive associations of ‘freedom’ and ‘self-direction’**
- 2. It is ‘age agnostic’ and could be achieved at any time**
- 3. ... so best to plan for it early so that you can achieve it early**
- 4. Sounds like you are free to continue doing paid work**
- 5. Sounds like you are free to continue doing unpaid work**

FIRE

FINANCIAL INDEPENDENCE RETIRE EARLY

Financial Independence Retire Early (FIRE)

Every expense should be compared to ...

... time spent at work to earn the purchase

Keep living expenses very low

... and aim for 'extreme saving' of 70% of after-tax income

Once investments are 30 times living expenses (roughly \$1 million)

... you can quit your day job and completely retire

Variations on FIRE

1. Fat FIRE

More traditional lifestyle who saves more than the average retirement investor

2. Lean FIRE

Minimalist living and extreme savings with a far more restricted lifestyle

3. Barista FIRE

Work casual job to cover some living expenses (so they don't erode investments too much)

4. Coast FIRE

Work part-time job but have enough saved to fund their current living expenses

Steven Kurutz (2018), 'How to Retire in Your 30s With \$1 Million in the Bank', *The New York Times*.

Alexandra Kerr (2024), 'Financial Independence, Retire Early (FIRE)' on Investopedia (website)

FIRE is interesting ... but remember ...

- 1. The nature of work can change as your career develops**
- 2. Being actively employed in meaningful work is helpful for happiness**
- 3. Saving 70% comes with significant sacrifices (relationships)**
- 4. If you are in a relationship ... they need to be '110% on-board'**
- 5. ... and isn't planning on having any children any time soon!**
- 6. Will people respect you if you don't work?**
- 7. Happiness is in the details ... so what will you do once you FIRE?**
- 8. ... and with whom?**

Financial Independence

A situation in which the income you are drawing from your investments is enough to cover your living expenses ... indefinitely!

... and it is a much better term to use than 'retirement'

How Much Is Needed?

Important activity (recap)

Think back to your values and the principles of money and happiness

Now imagine that you are age 60 and financially independent. Describe ...

1. Your closest relationship (romantic or otherwise)
2. Your relationships with your immediate family (children, parents, siblings)
3. Any paid or unpaid work to keep you engaged and serving others
4. Any hobbies or interests including with whom you practice them
5. Where you are living and with whom you are living
6. A typical holiday and with whom you are on holiday

Assuming you own your own dwelling and ignoring inflation ...

... what is the minimum income would you need to fund that lifestyle?

ASFA Retirement Standard

Annual living expenses assuming you are healthy and own your own home

... so no rent or home loan repayments

	Couple	Single
Comfortable lifestyle	\$72,663 p.a.	\$51,630 p.a.
Modest lifestyle	\$47,387 p.a.	\$32,915 p.a.

FIRE Rule of Thumb

Multiply by 30 to get investment required for financial independence ...

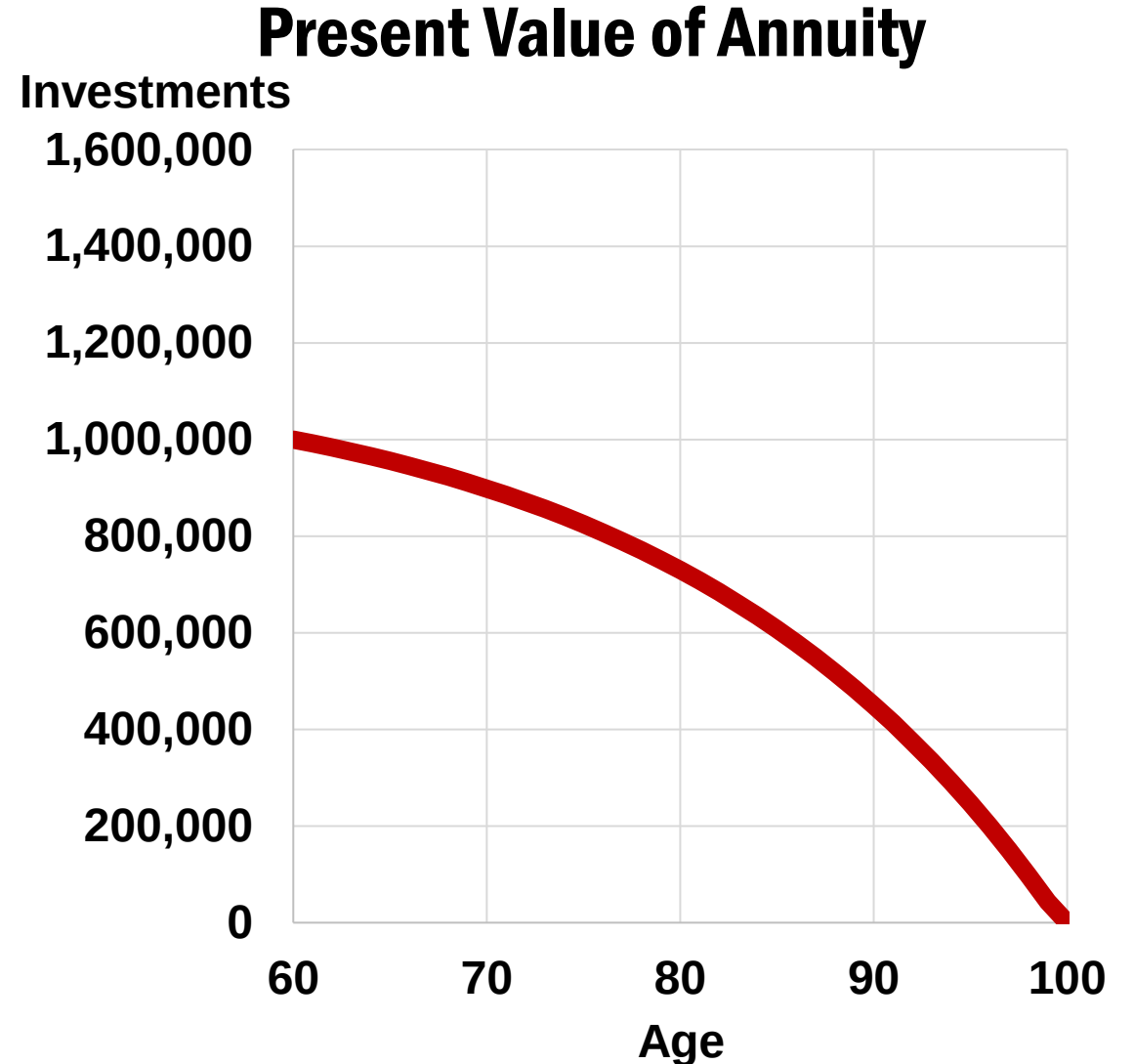
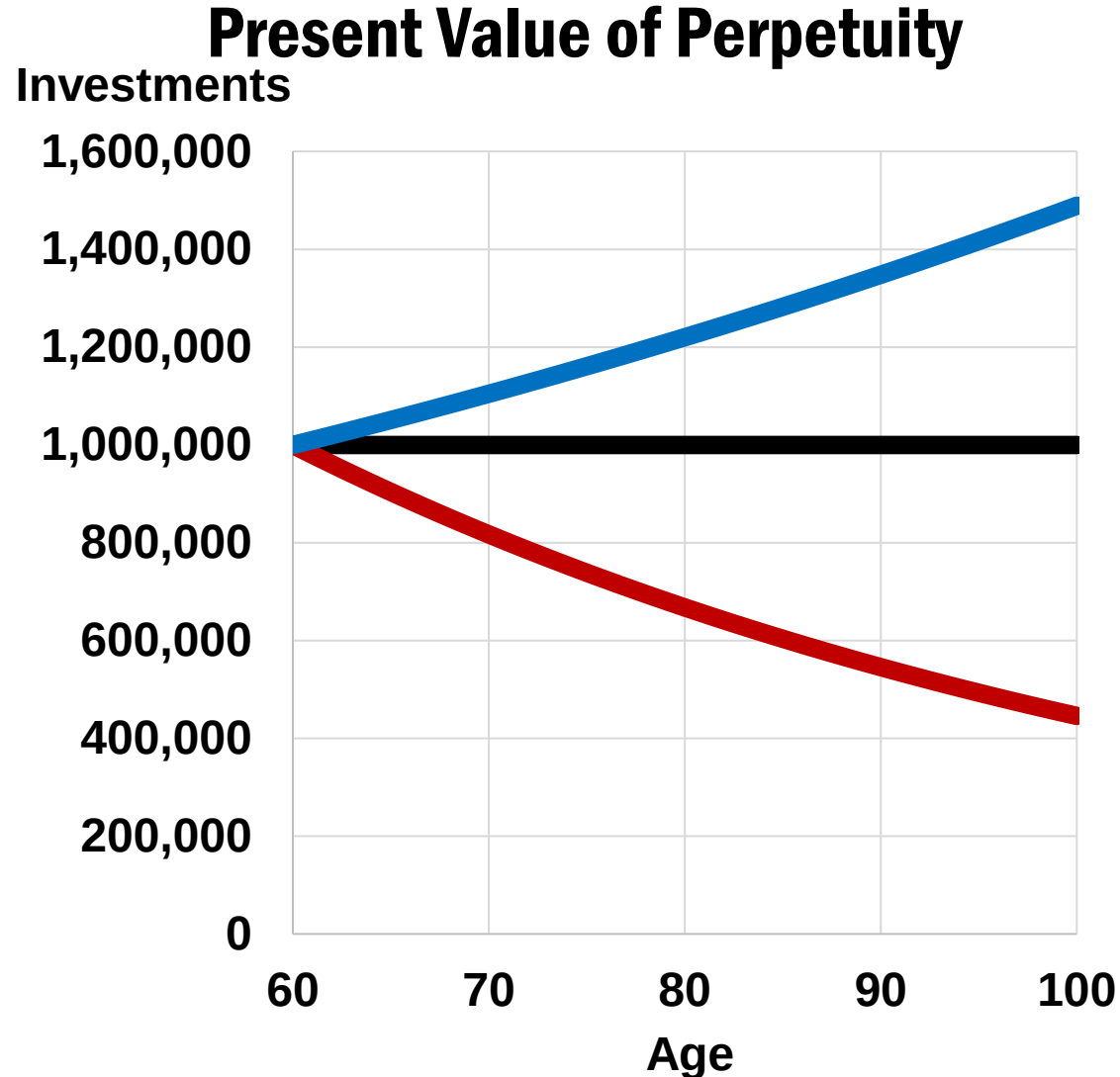
	Couple	Single
Comfortable lifestyle	\$2,179,890	\$1,548,900
Modest lifestyle	\$1,421,610	\$987,450

Steven Kurutz (2018), 'How to Retire in Your 30s With \$1 Million in the Bank', *The New York Times*.

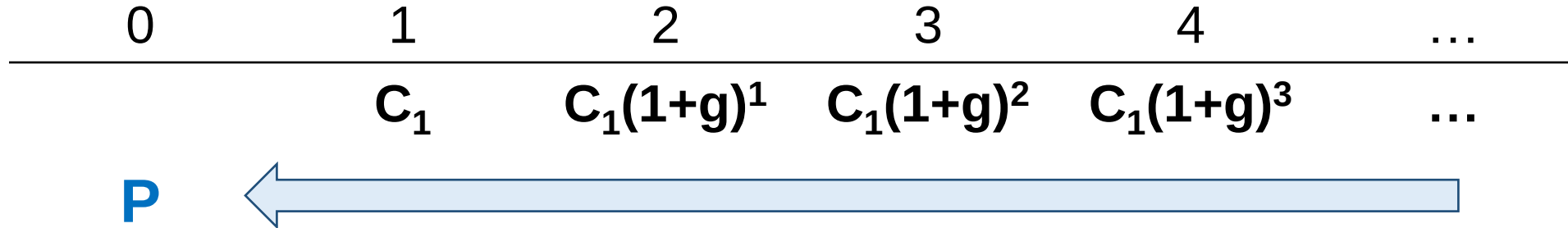
Alexandra Kerr (2024), 'Financial Independence, Retire Early (FIRE)' on Investopedia (website)

With math we can be more sophisticated!

Two mathematical approaches



What is a perpetuity?



A perpetuity is an infinite stream of regular cash flows

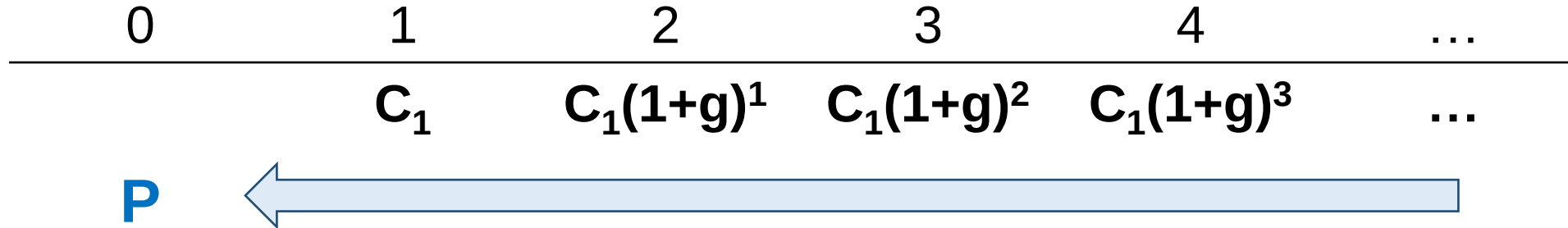
... the first cash flow occurs at the end of the first period (C_1)

... the cash flows can be level, growing or declining at a constant rate (g)

Useful for calculating the investment required

... to fund a perpetual stream of living expenses!

Present value of a growing perpetuity formula



$$P_{\infty,g} = \frac{C_1}{r - g} \quad g < r$$

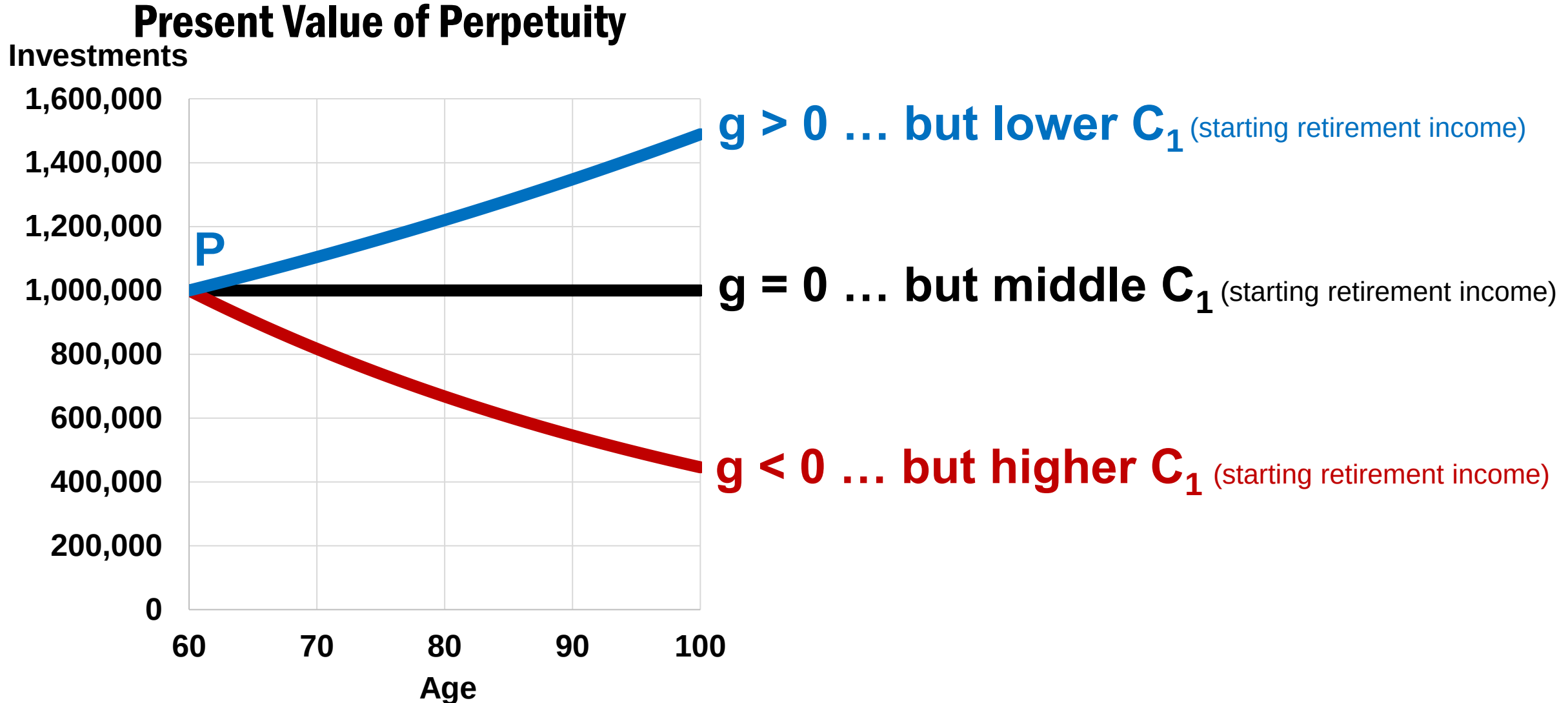
‘ C_1 ’ is the cash flow growing at a rate of ‘ g ’ per period

∞ means that they go on forever (with first cash flow at end of first period)

‘ r ’ is the effective rate of return per period (time value of money)

‘ P ’ is value of the infinite series of cash flows of ‘ C ’ at the start (left)

'g' is both growth in income and investments



Present value of constant perpetuity example

How much would Susan need for financial independence if she would like \$50,000 per year indefinitely, keeping pace with inflation ($g = 0$) if she can achieve real investment returns of 5% per annum?

$$P_{\infty, g} = \frac{C_1}{r - g}$$

Present value of constant perpetuity example

How much would Susan need for financial independence if she would like \$50,000 per year indefinitely, keeping pace with inflation ($g = 0$) if she can achieve real investment returns of 5% per annum?

$$P_{\infty, g} = \frac{C_1}{r - g}$$

$$= \frac{50,000}{0.05 - 0}$$

$$= \$1,000,000$$

Calculator

$$50000 \div 0.05 =$$

Spreadsheet

$$= 50000 / 0.05$$

Present value of growing perpetuity example

How much would Susan need for financial independence if she would like \$50,000 per year indefinitely, growing at 1% above inflation ($g = 1\%$) if she can achieve real investment returns of 5% per annum?

$$P_{\infty, g} = \frac{C_1}{r - g}$$

Present value of growing perpetuity example

How much would Susan need for financial independence if she would like \$50,000 per year indefinitely, growing at 1% above inflation ($g = 1\%$) if she can achieve real investment returns of 5% per annum?

$$P_{\infty, g} = \frac{C_1}{r - g}$$

$$= \frac{50,000}{0.05 - 0.01}$$

$$= \$1,250,000$$

Calculator

$$50,000 \div (0.05 - 0.01) =$$

Spreadsheet

$$= 50,000 / (0.05 - 0.01)$$

Present value of decreasing perpetuity example

How much would Susan need for financial independence if she would like \$50,000 per year indefinitely, growing at 1% below inflation ($g = -1\%$) if she can achieve real investment returns of 5% per annum?

$$P_{\infty, g} = \frac{C_1}{r - g}$$

Present value of decreasing perpetuity example

How much would Susan need for financial independence if she would like \$50,000 per year indefinitely, growing at 1% below inflation ($g = -1\%$) if she can achieve real investment returns of 5% per annum?

$$P_{\infty, g} = \frac{C_1}{r - g}$$

$$= \frac{50,000}{0.05 - -0.01}$$

$$= \$833,333$$

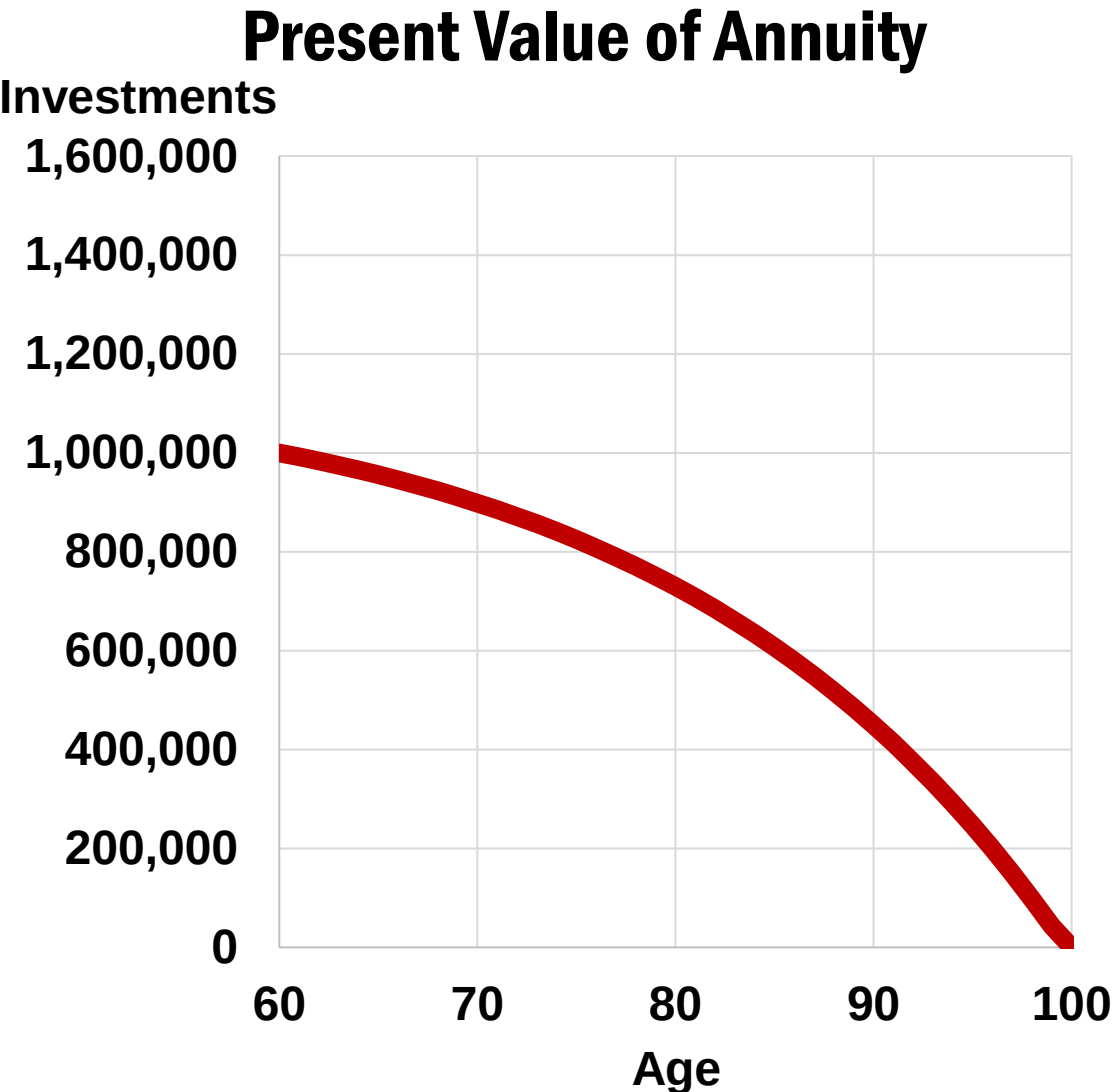
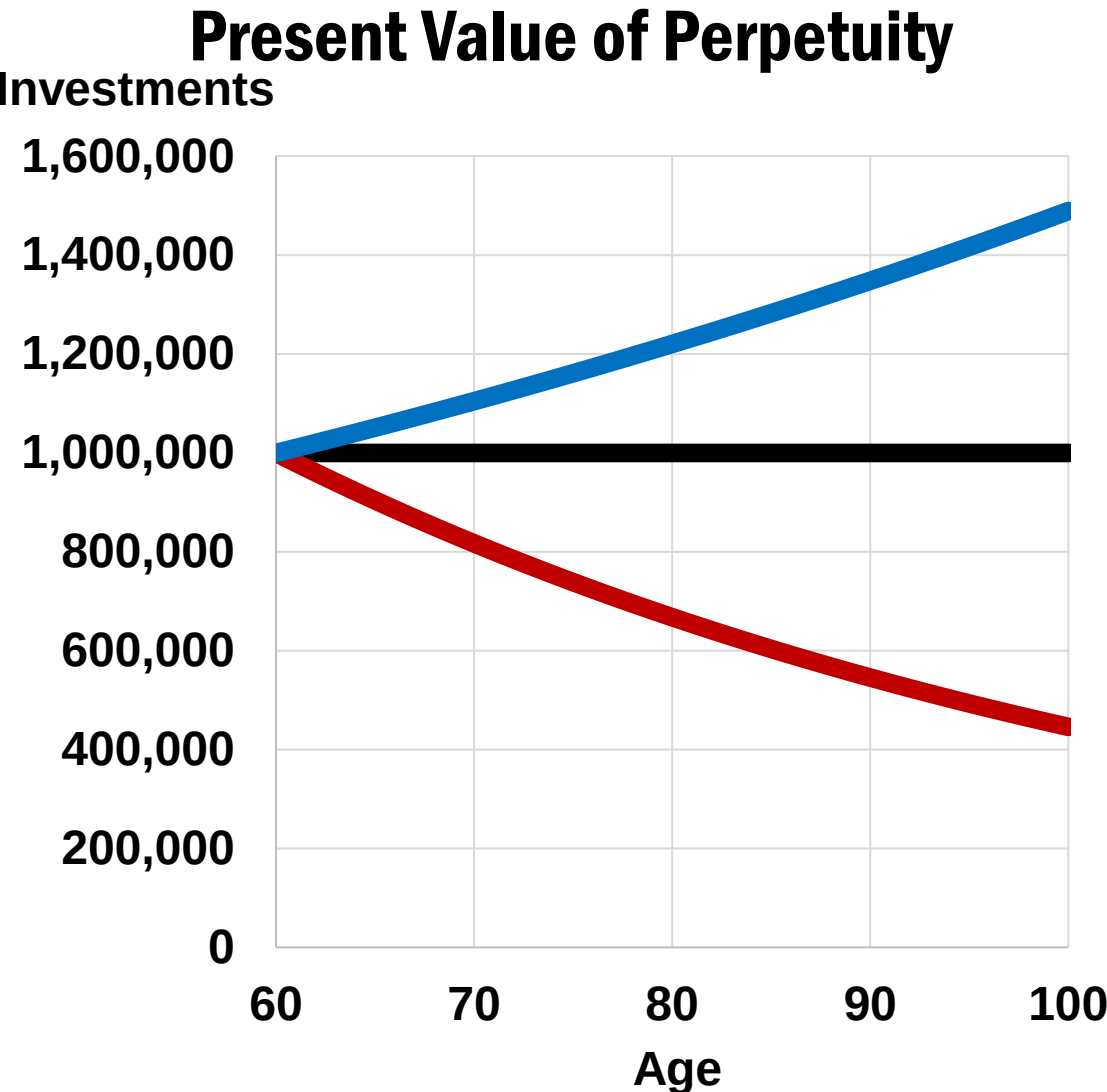
Calculator

$$50,000 \div (0.05 + 0.01) =$$

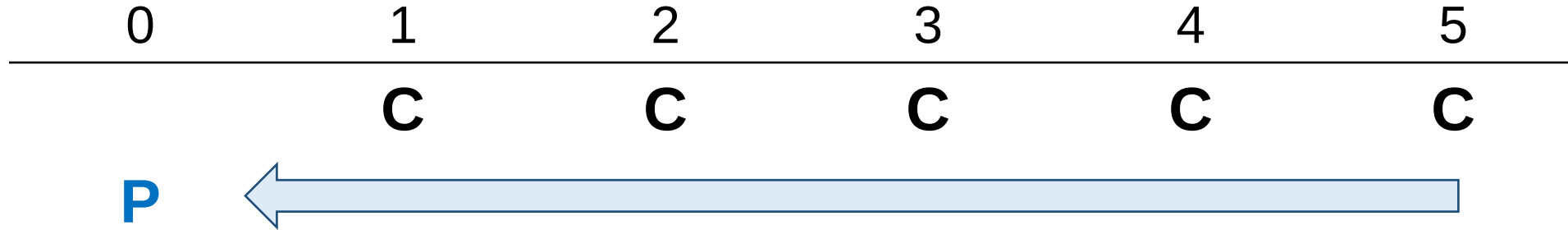
Spreadsheet

$$= 50,000 / (0.05 - -0.01)$$

What about the second approach?



Present value of an annuity



We are calculating the value of the cash flows (with first cash flow at end of first period)

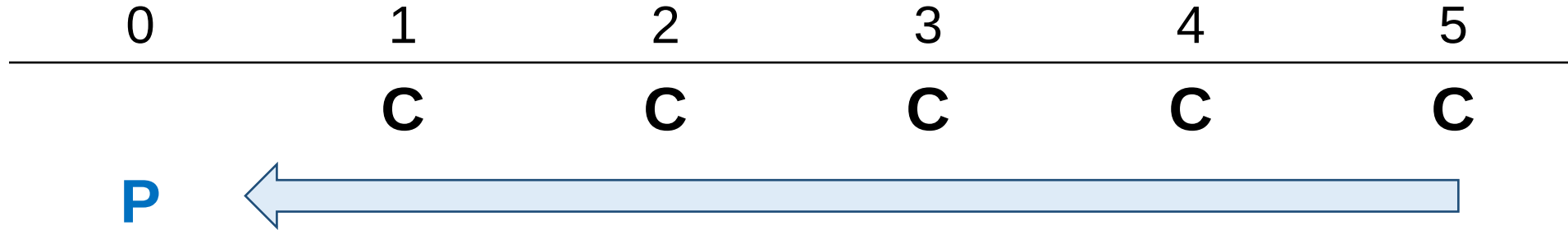
... at the **START** of the regular series of payments

Here the regular cash flow is 'C'

there are 5 cash flows and we are moving them to $t = 0$

to calculate the present value of the annuity 'P'

Present value of an annuity formula



$$P_n = C \times \left[\frac{1 - (1 + r)^{-n}}{r} \right]$$

‘C’ are the regular cash flows that are being moved to the left

‘n’ is the number of cash flows (with first cash flow at end of first period)

‘r’ is the effective rate of return per period (time value of money)

‘P’ is value of the regular series of ‘n’ cash flows of ‘C’ at the start (left)

Present value of annuity example

How much would Susan need for financial independence if she would like \$50,000 per year for 30 years, keeping pace with inflation, if she can achieve real investment returns of 5% per annum?

$$P_n = C \times \left[\frac{1 - (1 + r)^{-n}}{r} \right]$$

Present value of annuity example

How much would Susan need for financial independence if she would like \$50,000 per year for 30 years, keeping pace with inflation, if she can achieve real investment returns of 5% per annum?

$$P_n = C \times \left[\frac{1 - (1 + r)^{-n}}{r} \right]$$

$$= 50,000 \times \left[\frac{1 - 1.05^{-30}}{0.05} \right]$$

Calculator

$$50000 \times (1 - 1.05^{-30}) \div 0.05 =$$

Spreadsheet

$$= 50000 * (1 - 1.05^{-30}) / 0.05$$

= **\$768,623** compared to \$1,000,000 for a constant perpetuity (indefinitely)

Asset categories real and nominal returns

	Asset category	Real return		Inflation		Nominal return
1.	Cash	0.5%	+	2.5%	=	3.0%
2.	Fixed Interest	1.5%	+	2.5%	=	4.0%
3.	Residential Property	5.0%	+	2.5%	=	7.5%
4.	Commercial Property	6.0%	+	2.5%	=	8.5%
5.	Australian Shares	6.0%	+	2.5%	=	8.5%
6.	International Shares	6.0%	+	2.5%	=	8.5%

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10.

Asset categories price gains and income

	Asset category	Price gains		Income		Nominal return
1.	Cash	0.0%	+	3.0%	=	3.0%
2.	Fixed Interest	0.0%	+	4.0%	=	4.0%
3.	Residential Property	5.0%	+	2.5%	=	7.5%
4.	Commercial Property	5.0%	+	3.5%	=	8.5%
5.	Australian Shares	5.0%	+	3.5%	=	8.5%
6.	International Shares	6.5%	+	2.0%	=	8.5%

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10.

How Can I Achieve It?

Modelling
a path to financial
independence

To achieve Financial Independence ...

1. Calculate financial independence goal

\$PV of future living expenses at the financial independence target age

2. Calculate future value of existing savings

FV of single cash flow at target date

3. Calculate financial independence gap*

Financial independence goal from step 1 minus FV of existing savings from step 2

4. Calculate regular savings required

Future value of annuity formula solving for 'payment' C with gap from step 3 as the target

... then implement a plan to achieve this!

* Receiving an inheritance might help to close this gap but we'll assume zero inheritance in this example

Financial Independence Example

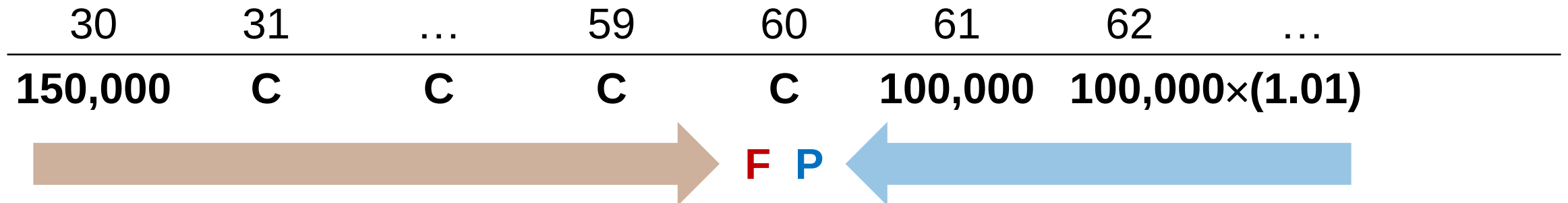
Current Age: 30

Independence Age: 60

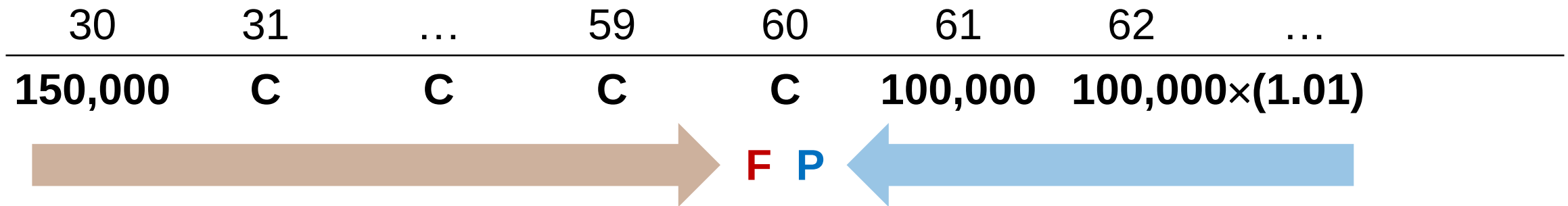
Future Expenses: \$100,000 growing at 1% p.a. in real terms

Current Savings: \$150,000 invested at 6% p.a. in real terms

How much to save each year to achieve financial independence?



1. Calculate Financial Independence Goal



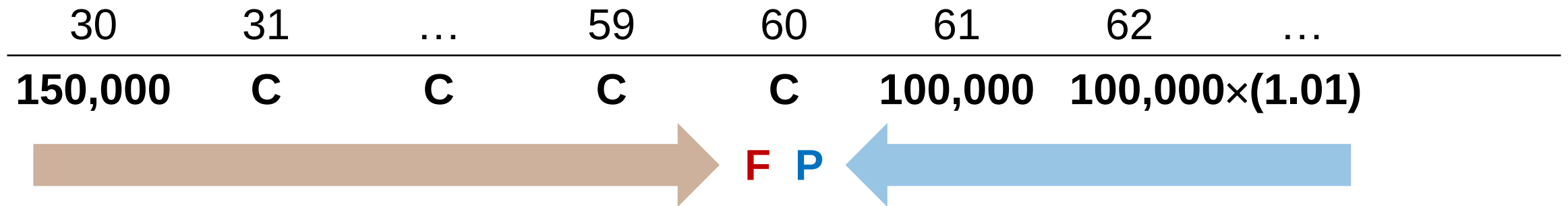
We are calculating the amount required at age 60 (P_{60})

To fund expenses starting at age 61* ($C_{61} = \$100,000$) growing at 1% p.a.
assuming our investments achieve a real return of 6% p.a.

$$P_{60,\infty,g} = \frac{C_{61}}{r - g} = \frac{100,000}{0.06 - 0.01} = \$2,000,000$$

* You can model this question using monthly cash flows but to make it more realistic but it doesn't make a big difference

2. Calculate Future Value of Existing Savings



We are calculating the amount we expect to have at age 60 (F_{60})

based on only our existing savings of \$150,000

assuming our investments achieve a real return of 6% p.a.

$$F_{60} = P \times (1 + r)^n = 150,000 \times 1.06^{30} = \$861,524$$

3. Calculate Financial Independence Gap*

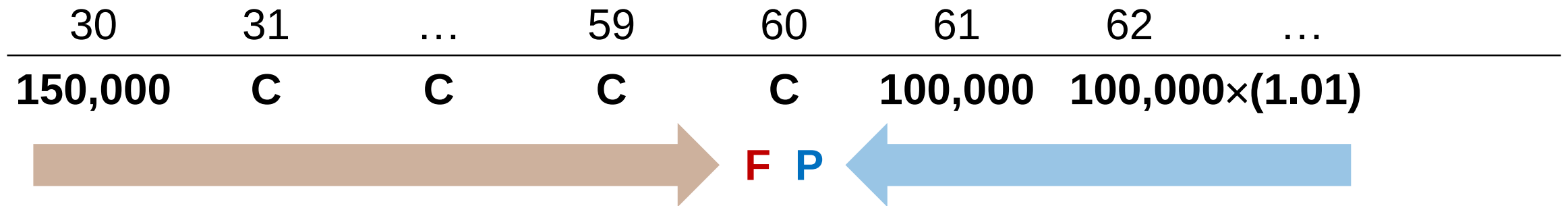
30	31	...	59	60	61	62	...
150,000	C	C	C	C	100,000	100,000×(1.01)	
				F P			

We are calculating gap (G_{60}) between how much we need (P_{60})
... and how much we expect to have (F_{60}) based on existing savings

$$\begin{aligned} G_{60} &= P_{60} - F_{60} \\ &= 2,000,000 - 861,524 \\ &= \$1,138,476 \end{aligned}$$

* Receiving an inheritance might help to close this gap but we'll assume zero inheritance in this example

4. Calculate Regular Savings Required



We regular savings required (C) to close the gap at age 60 of \$1,138,476 assuming our investments achieve a real return of 6% p.a.

$$C = F_n \div \left[\frac{(1 + r)^n - 1}{r} \right] = 1,138,476 \div \left[\frac{1.06^{30} - 1}{0.06} \right]$$

$$= \$14,400 \text{ per annum}$$

To achieve Financial Independence ...

1. Calculate financial independence goal

\$PV of future living expenses at the financial independence target age

2. Calculate future value of existing savings

FV of single cash flow at target date

3. Calculate financial independence gap*

Financial independence goal from step 1 minus FV of existing savings from step 2

4. Calculate regular savings required

Future value of annuity formula solving for 'payment' C with gap from step 3 as the target

... then implement a plan to achieve this!

* Receiving an inheritance might help to close this gap but we'll assume zero inheritance in this example

What Are The Common Strategies?

3 Key Principles

- 1. Consistently 'save to invest' at least 20% of your income (before-tax)**
- 2. Avoid dumb investment decisions**
- 3. Diversify across multiple financial strategies**

1. Invest in your ability to generate income

Your brain is your most valuable asset right now

If it can generate a graduate income of just \$50,000 for 40 years at a required return of 5% per annum then it is worth:

$$P_n = C \times \left[\frac{1 - (1 + r)^{-n}}{r} \right] = 50,000 \times \left[\frac{1 - 1.05^{-40}}{0.05} \right] = \$857,954$$

Invest time into your career plan (next Unit)

Invest in ongoing work-relevant education

Invest in your physical and mental health

2. Invest in a minimum of one property*

Controls dwelling expenses over the long-term

Rent increases with average income and inflation ... but mortgage payments do not

Expected price growth linked to growth in average income

Real price growth 2.5% + Inflation 2.5% = Nominal price growth 5.0% per annum

You also save on paying rent of about 3% of property price

May only need a 2 bedroom apartment or small house when you are 60+

There are benefits from downsizing from large house and investing the difference

Property can be used as security on investment loans

... which gives you the option to invest in other assets (such as property* or shares)

* Buying more than one property can add to the problem of housing affordability (unless you are buying a newly constructed home) and so later in this course we will discuss how to invest in shares rather than buying an investment property.

3. Invest in a diversified portfolio of shares

Provides growth over long-term

... plus regular income through company profits distributed as dividends

Expected price growth linked to average income

Real price growth 2.5% + Inflation 2.5% = Nominal price growth 5.0% per annum

You also receive dividends that average about 3.5% per year

Invest in Exchange Traded Funds

These are diversified across 100 or more companies

Minimum investment time horizon is 5 years

Buy and hold over the long-term

Avoid short-term speculation (gambling)

Avoid platforms that like eToro. Avoid speculating on cryptocurrency, forex or commodities

4. Use your retirement savings effectively

Retirement savings systems have tax advantages

Concessional (low) tax on money going into the system

Concessional (low) tax on investment returns

Tax-free income streams after the age of 60 in Australia

5. Get involved in a start-up or start your own business

Start-ups can generate millions of dollars of wealth in just a few years

They also provide valuable business lessons whether they succeed or fail

Avoid investing a significant amount of your own money

Best to join as an early staff member and negotiate a generous employee stock option plan (ESOP) in exchange for a basic salary ‘sweat equity’

‘Do your time’ for a few years then either switch to another startup if it fails or go back to a large organization

... or start your own business as the founder

(preferably after at least 5 years of experience in the industry)

Remember to diversify

- 1. Invest in your ability to generate income**
- 2. Invest in at least one property**
- 3. Invest in a diversified portfolio of shares**
- 4. Use your retirement savings effectively**
- 5. Get involved in a start-up** (or start your own business)

Don't rely on only one or two strategies

Best to diversify between three or four strategies

The Superannuation System

Superannuation

Retirement savings system in Australia is called 'superannuation'

'Superannuate' means to retire with a pension (regular income stream)

Retirement savings systems elsewhere are usually just called 'pension system'

You usually cannot access your superannuation until you turn 60

You cannot use it to buy a new car or to go on holiday before age 60!

Your employer makes regular contributions

Those contributions are usually taxed at 15%

... rather than marginal income tax rate + Medicare levy (34.5% for most people)

... which are invested and accumulate over 30+ years

... at which point you start drawing a regular income from your super

‘Superannuation’ or ‘Super’

Most professionals use the full term ‘superannuation’

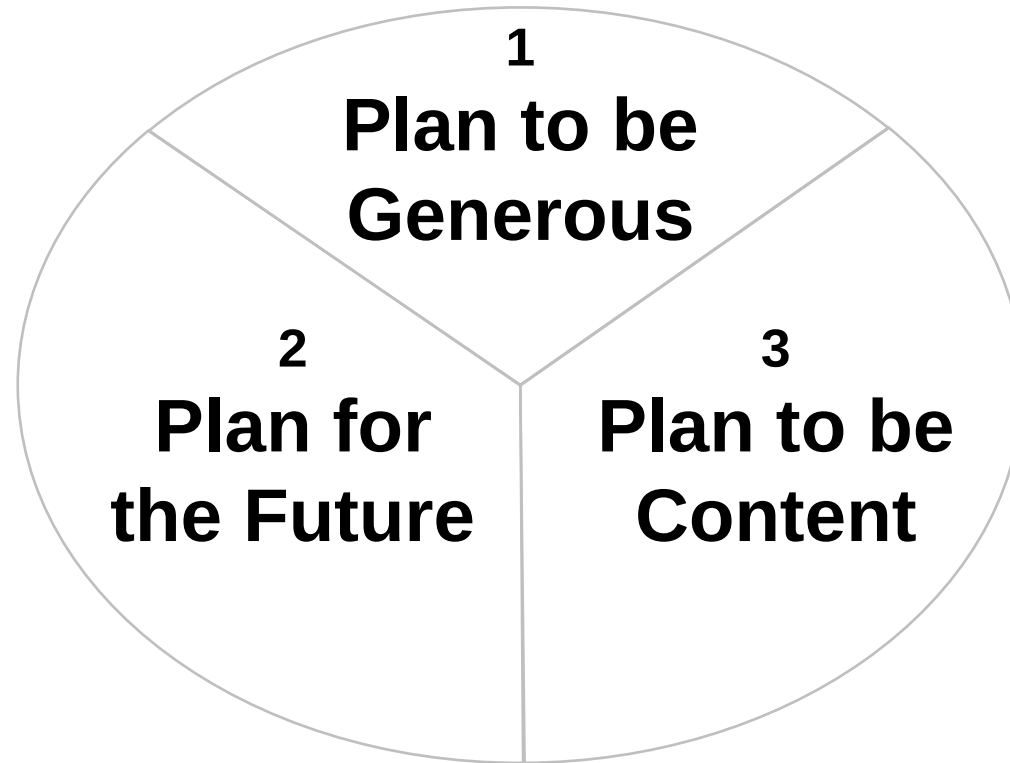
Calling it ‘super’ is a bit colloquial

I’m going to use ‘super’ from this point for brevity

Super and the Pie

Your **pie** is your
total income
each month

and what you do with it!



Investment income covers all three

Two types of systems

1. **Accumulation** (also 'defined contribution')

Account balance is sum of past contributions + returns

Retirement income depends on account balance

Risk of underfunded retirement is borne by the individual

Risk of default is very low

2. **Defined Benefit** (this is being phased out in Australia)

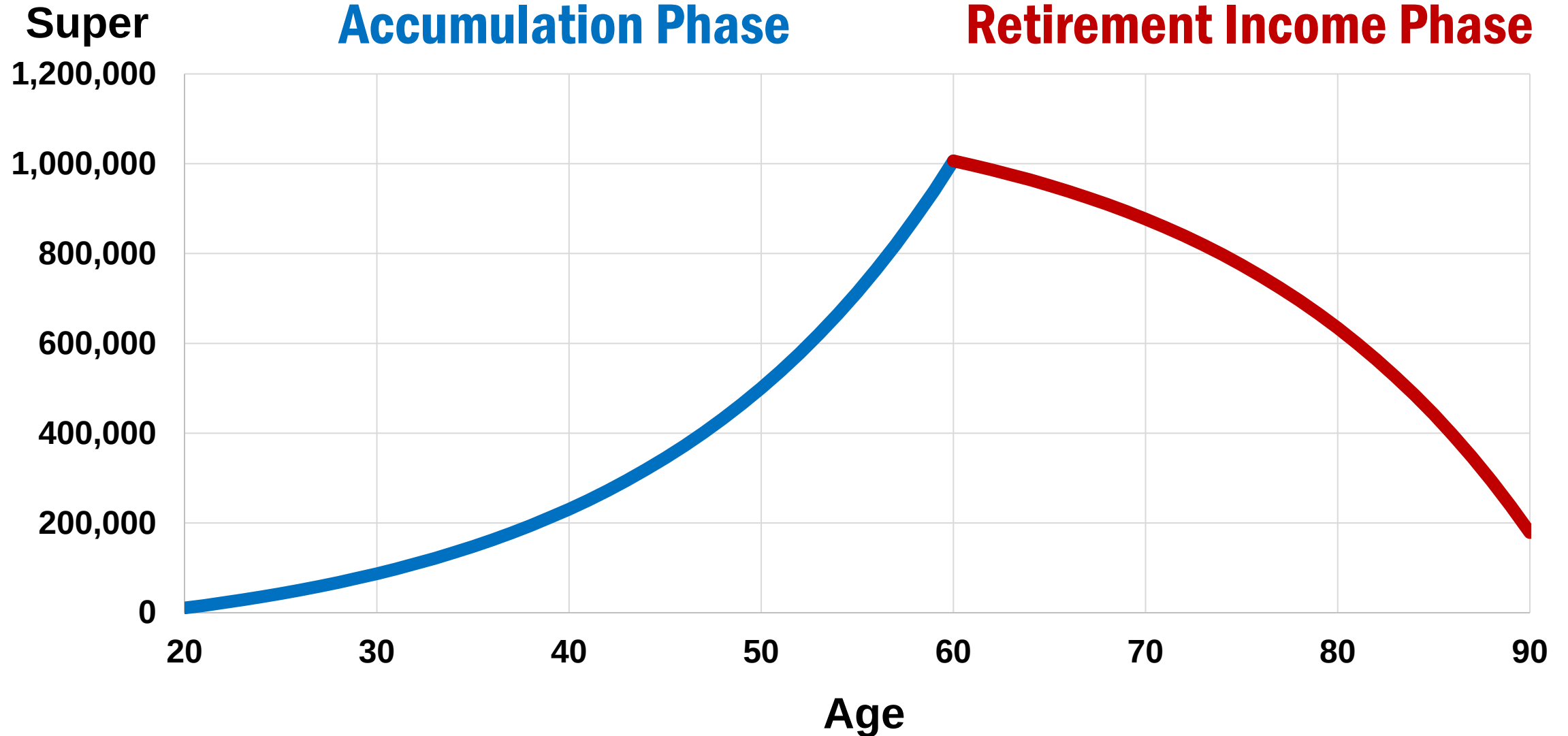
Retirement income depends on salary in final years before retirement

Guaranteed income stream until death (with partial reversion to spouse)

Risk of underfunded retirement is borne by the company (or government)

Risk of default is medium to high (often needs to be underwritten by government)

Two key phases



How much can be accumulated in super?

There is a limit to how much you can accumulate in super

... and still receive concessional tax treatment

The limit is called the ‘transfer balance cap’

Lifetime limit that can be transferred from accumulation phase to retirement income phase

Transfer Balance Cap: \$1.9 million per person

Indexed in \$100,000 increments every few years in line with CPI

This is ‘per individual’ so a couple can accumulate double this if equally balanced

Anything above this is called an ‘Excess Transfer’

These can be subject to an excess transfer balance tax of 15% (first) to 30% (second +)

Types of funds

1. MySuper

Low fees, simple features, few investment options. Offered by many retail and industry funds.

2. Industry super funds

Low to medium fees offered by not-for-profit organisations (some linked to unions)

3. Retail super funds

Medium to high fees offered by financial institutions. More investment choices.

4. Public sector super funds

For government employees

5. Corporate super funds

For employees of large companies such as Telstra and Qantas.

6. Wrap accounts

Popular with financial advisers as they offer sophisticated investment options.

7. Self-managed super funds

Fund set up for a family that maintains direct control over assets. High accounting fees.

Life insurance within super

Death cover

Lump-sum payment upon death of member

Total and Permanent Disability (TPD)

Lump-sum payment upon total and permanent disability

Salary continuance (income protection)

Replaces part of income if accident or illness prevents you from working

Insurance premiums are quite cheap within super

Need to be careful don't lose benefits when changing funds

Premiums can be a drag on achieving long-term goals

When can I access my super?

The earliest age you can access super is called your 'preservation age'

The preservation age is currently age 60

Exceptions:

- First Home Super Saver Scheme

- Compassionate Grounds

- Financial Hardship

- Emergency Release Programs (such as COVID-19 in 2020)

What if your super fund goes bankrupt?

Unlikely to happen because super funds are closely regulated by APRA.

Superannuation investments are held in a 'trust' which is completely separate from the financial institution's own accounts and assets.

These 'trusts' do not have any significant debts or liabilities so it is not really possible for them to go bankrupt.

If financial institution goes bankrupt then trust remains separate and liquidator will sell off trust to another superannuation fund to administer.

Bottom line – DON'T WORRY!

The Good

- 1. Regular savings paid for you by employer**
- 2. Concessional (low) tax on money going into super**
- 3. Concessional (low) tax on investment returns while inside super**
- 4. Concessional (low) tax when drawing income from super**
- 5. Good investment options with good returns**
- 6. You normally can't access your super until at least age 60**
- 7. The Australian Government supports the system**

The Bad

- 1. Super rules are quite complicated and confusing**
- 2. Many different super funds resulting in 'choice anxiety'**
- 3. Easy to accumulate multiple super funds with changing jobs**
- 4. Some super funds have high fees and poor returns**
- 5. Some funds give you default life insurance when you don't need it**
- 6. You normally can't access your super until at least age 60**
- 7. The Government keeps changing the rules ... resulting in anxiety**

Break

Consolidating Your Super

Multiple super accounts is not good

Duplicated account fees

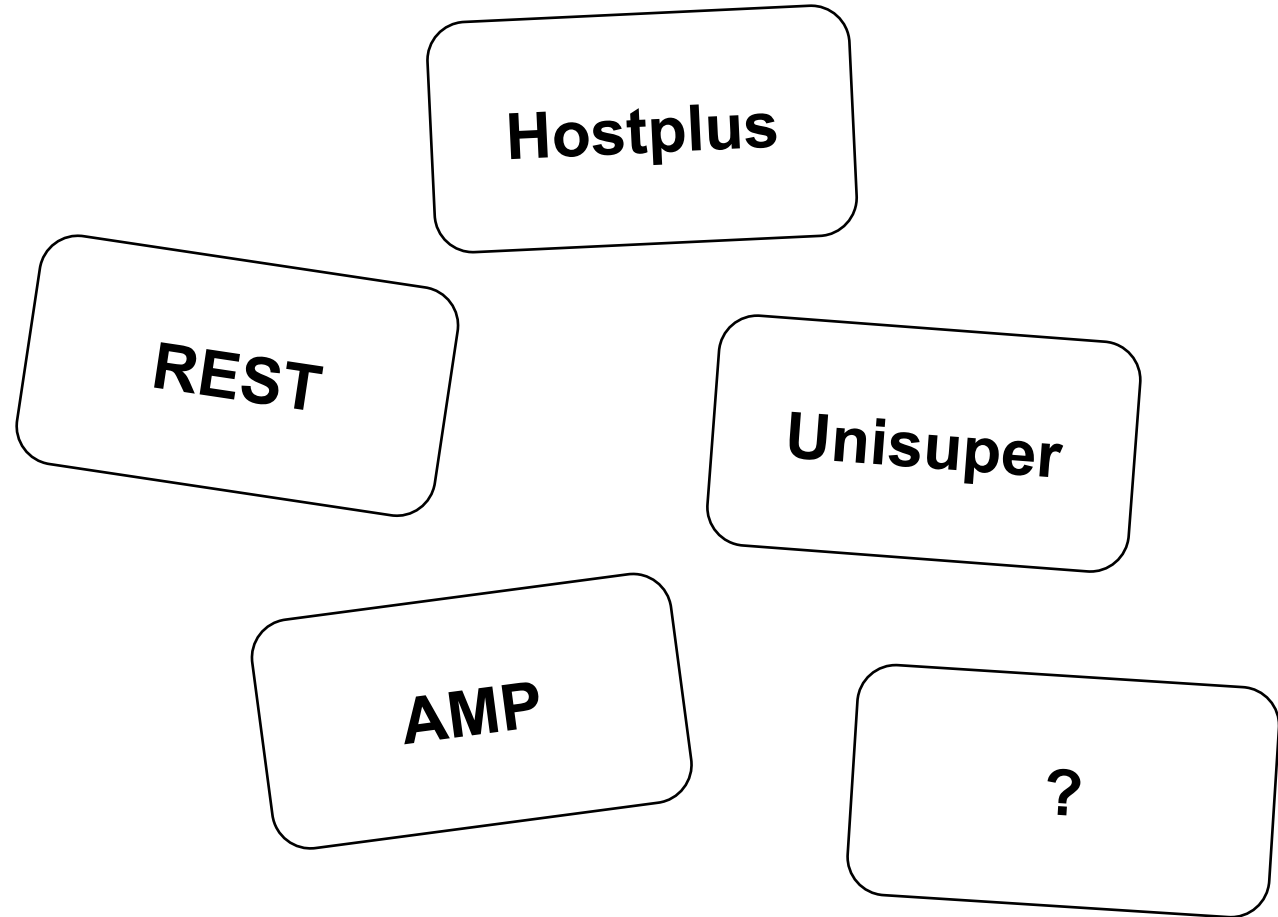
Duplicated insurance

Duplicated premium

Duplicated paperwork

Risk of losing track of it

Stress from disorganisation



Super Stapling ... commenced Nov 2021

**You are 'stapled' to the first fund you sign up with
... or the fund you are with now.**

**So when you move jobs your super fund will 'move' with you
... unless you choose a different super fund.**

**You are not 'stuck' with the same super fund.
You can change fund whenever you like.**

Consolidating multiple accounts into one

1. Go to my.gov.au
2. Log in or create an account
3. Link your myGov account to the ATO
4. Select 'Super' and then 'Manage'
5. Select 'Transfer super'

This option will only appear if you have more than one super account

When closing an account, watch out for ...

1. Is an employer still making contributions into the old fund?

A few (not many) employers pay higher contributions into selected funds

Closing a fund that receives active employer contributions causes problems

Some employers don't allow 'choice of fund'

2. Will you lose insurance cover that cannot be replicated?

You may now have a 'pre-existing medical condition' that means you could lose cover

3. Bad behaviour from some 'industry funds'

If they don't have a signature on file or the name is slightly different they can be problematic

It can be a good idea to call your old funds before closing the account!

Searching for lost super

Superannuation funds are required to report ‘lost members’

They have not been able to contact you

They have not received any contributions or rollovers for 5 years

Your account was transferred from another fund as a lost member

The ATO keeps a ‘Lost members register’

- 1. Search online via MyGov (manage my super)**
- 2. Call ‘Lost super search line’ on 13 28 65**
- 3. Submit a ‘Searching for lost super’ form**

What jobs may not have paid super?

You were under 18 and worked less than 30 hours per week

... unless covered by an employer agreement (or your employer was generous!)

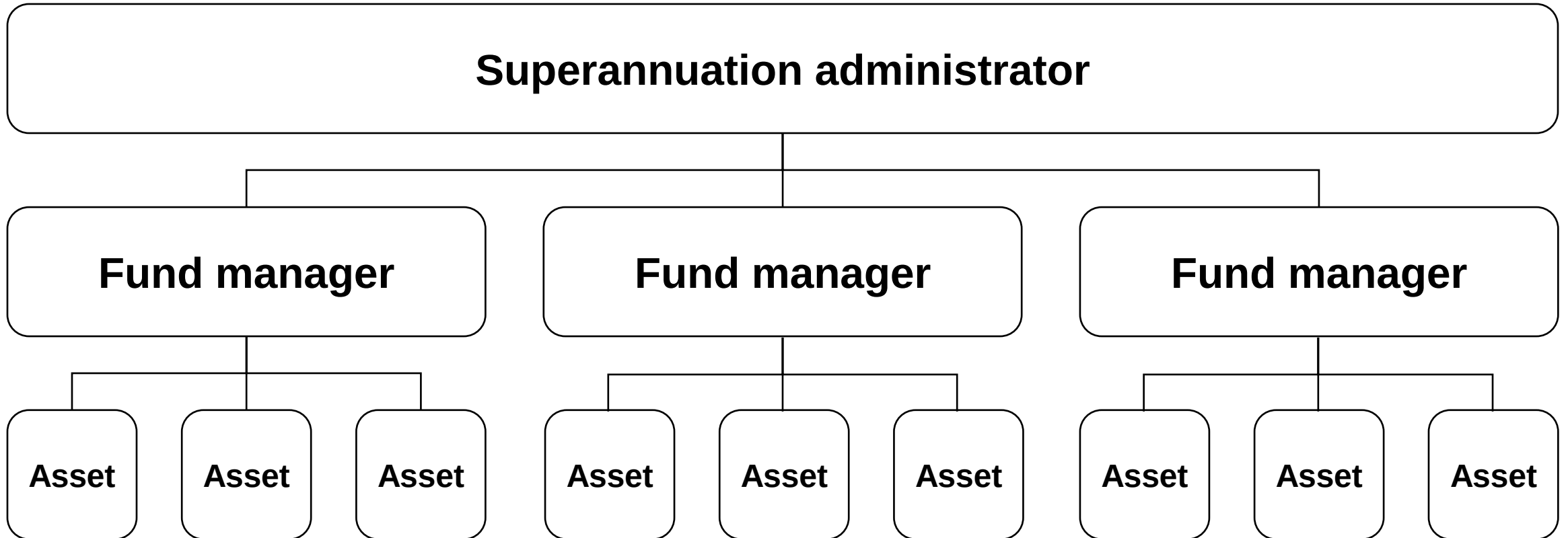
You were a ‘domestic worker’ and worked less than 30 hours per week

Examples of private domestic workers include a nanny, cleaner or gardener

Self-employed work (when you are not an employee of the company)

Investment Options

Three layers



Investment alternatives

1. Investment options

Super fund chooses both asset allocation and managed funds

You choose between 'conservative', 'balanced', 'growth' ...

Low fees but reduced choice and flexibility

2. Asset allocation

You choose asset allocation and super fund chooses the fund managers

You choose allocation between cash, fixed interest, shares, listed property ...

Slightly higher fees in exchange for more choice and flexibility

3. Direct investment

You choose both asset allocation and fund managers (and sometimes specific shares)

You choose specific manage funds (or shares) from a large menu of options

Higher fees in exchange for maximum choice and flexibility

Should I invest in low-risk investments?

... these funds are supposed to fund retirement after all!

In Units 9 and 10 we will define 'total risk' as the volatility of returns

We will see that there is a trade-off between total risk and expected returns

Better to choose higher total risk below the age of 50

... and then gradually reduce investment risk levels after that

Asset categories

Asset category	Asset category	
1. Cash	Transaction, Savings Account or CMT	Lower risk
2. Fixed Interest	Term deposits, debentures or bonds	Short time-horizon
3. Residential Property	Apartment, townhouse, duplex or house	
4. Commercial Property	Listed Property Trust	
5. Australian Shares	Australian Exchange Traded Fund (ETF)	Higher risk
6. International Shares	Global Exchange Traded Fund (ETF)	Long time-horizon

Cryptocurrency (bitcoin), commodities (gold) and foreign exchange are speculative investments

CMT is a Cash Management Trust. These are offered by investment banks such as Macquarie Bank and are popular with financial advisers
A 'speculative investment' is a euphemism for 'gambling'. This is because they generate no cash flows. More on this in Units 9 and 10.

Asset categories

Asset category	Asset category	Time horizon
1. Cash	Transaction, Savings Account or CMT	0 – 1 years
2. Fixed Interest	Term deposits, debentures or bonds	1 – 5 years
3. Residential Property	Apartment, townhouse, duplex or house	5+ years
4. Commercial Property	Listed Property Trust	5+ years
5. Australian Shares	Australian Exchange Traded Fund (ETF)	5+ years
6. International Shares	Global Exchange Traded Fund (ETF)	5+ years

Cryptocurrency (bitcoin), commodities (gold) and foreign exchange are speculative investments

CMT is a Cash Management Trust. These are offered by investment banks such as Macquarie Bank and are popular with financial advisers
A 'speculative investment' is a euphemism for 'gambling'. This is because they generate no cash flows. More on this in Units 9 and 10.

Past returns are no indicator of future returns

Many investors look to the past to extrapolate to the future.

We will see in Units 9 and 10 that this is a very bad idea.

**Expected future returns should be based on asset categories
... and not historical returns.**

Asset categories real and nominal returns

	Asset category	Real return		Inflation		Nominal return
1.	Cash	0.5%	+	2.5%	=	3.0%
2.	Fixed Interest	1.5%	+	2.5%	=	4.0%
3.	Residential Property	5.0%	+	2.5%	=	7.5%
4.	Commercial Property	6.0%	+	2.5%	=	8.5%
5.	Australian Shares	6.0%	+	2.5%	=	8.5%
6.	International Shares	6.0%	+	2.5%	=	8.5%

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10.

Asset categories price gains and income

	Asset category	Price gains		Income		Nominal return
1.	Cash	0.0%	+	3.0%	=	3.0%
2.	Fixed Interest	0.0%	+	4.0%	=	4.0%
3.	Residential Property	5.0%	+	2.5%	=	7.5%
4.	Commercial Property	5.0%	+	3.5%	=	8.5%
5.	Australian Shares	5.0%	+	3.5%	=	8.5%
6.	International Shares	6.5%	+	2.0%	=	8.5%

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10.

Financial advisers like Wrap Accounts

and also 'Managed Discretionary Accounts (MDAs)'

... such as Macquarie Wrap

Provide many investment choices

Allowing tailored asset allocation and selection of assets to suit the client

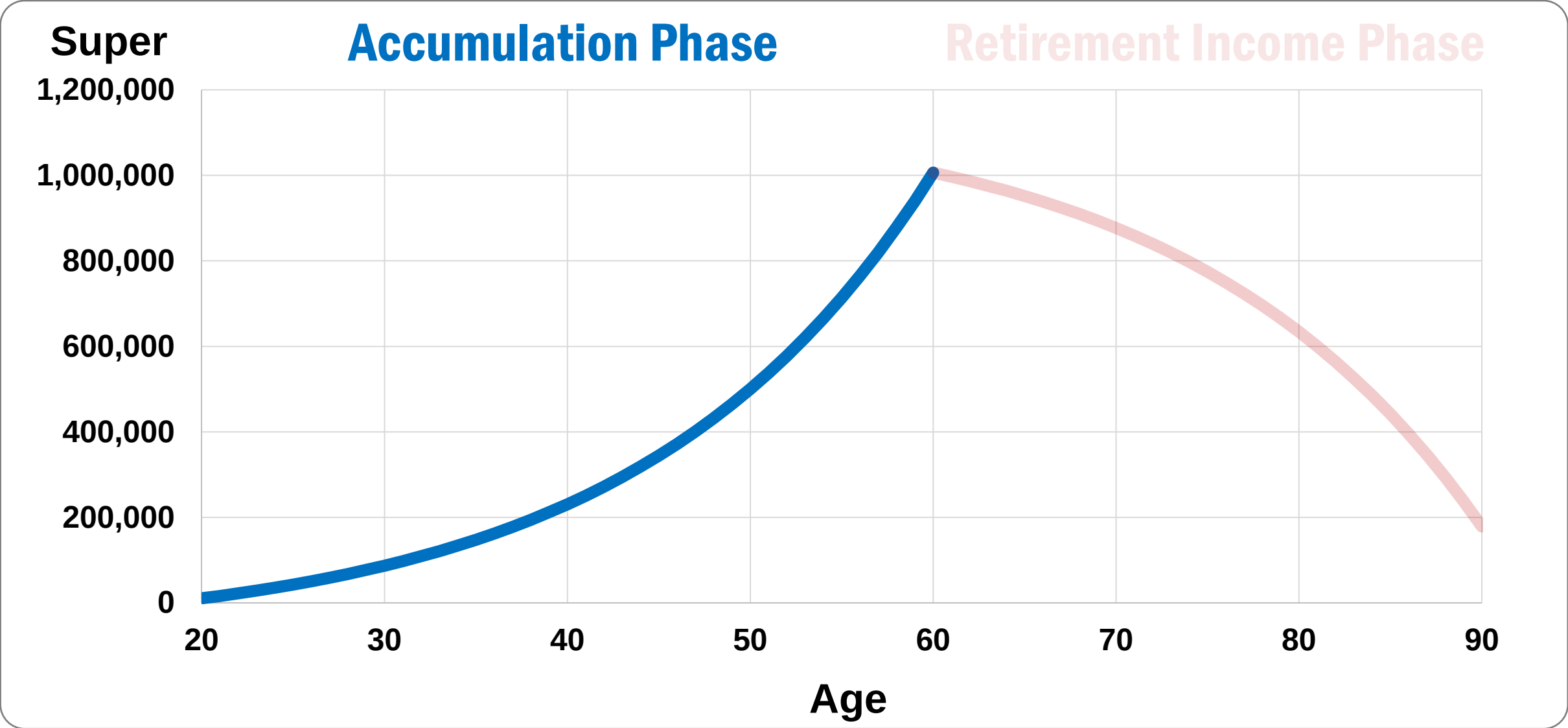
Adviser fees can be deducted directly from investment returns

Easy administration and reporting across many clients

... at a cost of higher administration fees paid by the client.

Making Contributions Into Super

Focussing on the accumulation phase here ...



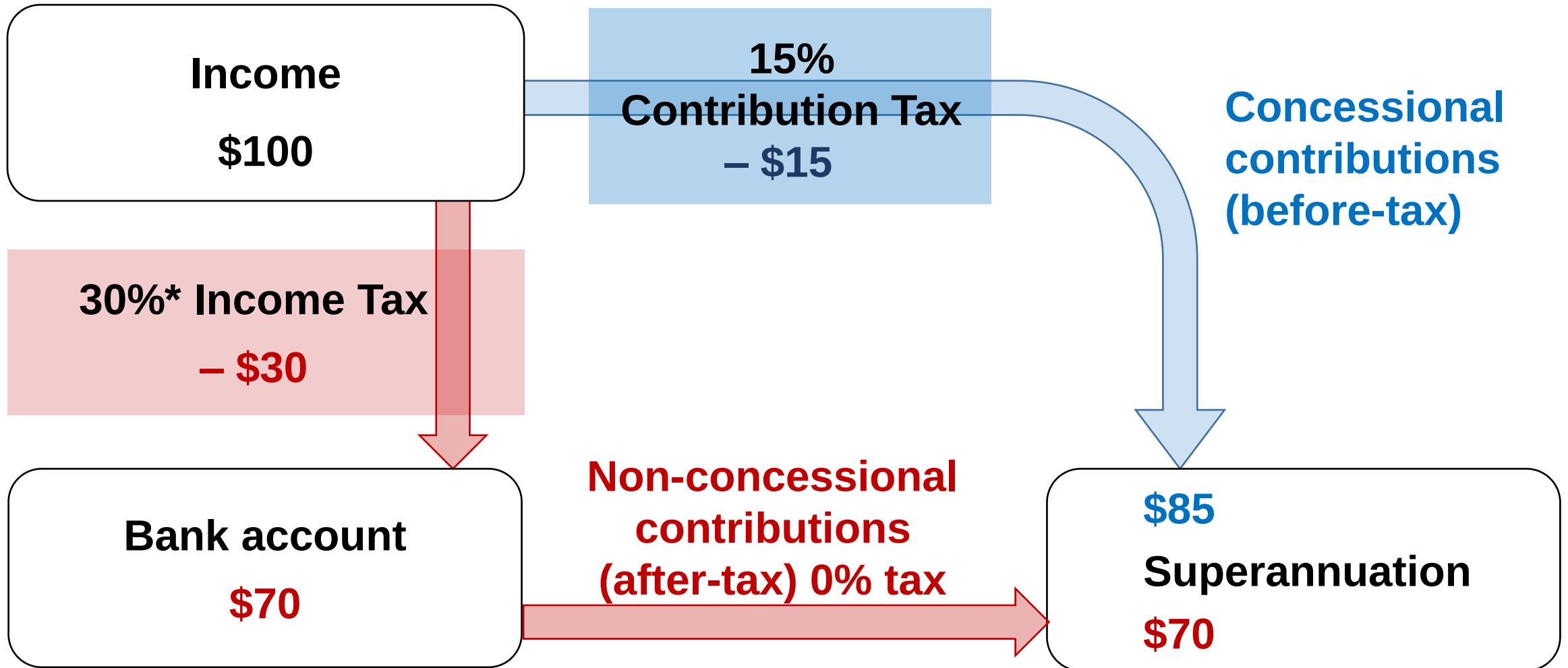
Income tax rates don't help with saving!

Taxable Income	Marginal Tax Rate
0 – \$18,200	Nil
\$18,201 – \$45,000	16%
\$45,001 – \$135,000	30%
\$135,001 – \$190,000	37%
\$190,001 and over	45%

Let's assume
a marginal tax
rate of
30%

The above rates do not include the Medicare levy of 2%

Contributions visualised



* The above Income Tax rate does not include 2% Medicare Levy (for simplicity). There is no Medicare Levy on Contribution Tax.

Limits for **concessional** contributions

These are 'before income-tax' contributions

Including employer Super Guarantee contributions or salary sacrifice contributions

Contributions tax rate is 15% rather than paying marginal tax rate plus Medicare Levy

Concessional Contribution Cap (maximum): \$30,000 per year

Any contributions above the cap taxed at marginal income tax rate plus Medicare Levy

Can carry-forward unused cap over 5 year rolling period if super balance < \$500k

High income earners may pay an additional Division 293 Tax of 15%

The Division 293 threshold is \$250,000 p.a. based on combined income and contributions

This is still lower than the highest marginal tax rate of 47% (including Medicare Levy)

Personal after-tax contributions (with deductions) limited if aged 67+

Age 67 to 74: must work 40 hours in consecutive 30 day period. Age 75+: not allowed.

Employer and salary sacrifice contributions still allowed until age 75 with no work test.

Australian Taxation Office (2024), 'Contributions caps' on ato.gov.au (website)

Australian Taxation Office (2024), 'Division 293 tax on concessional contributions by high-income earners' on ato.gov.au (website)

Limits for **non-concessional** contributions

These are '**after** income tax' contributions made from bank account

Including funds released by selling other investments and moving the funds into super

Super account balance must be below Transfer Balance Cap (\$1.9 million)

Non-concessional Contribution Cap (maximum): **\$120,000 per year**

Age restrictions:

Age < 75: may be able to 'bring forward' 2 years of caps and contribute up to \$360,000

Age ≥ 75: not allowed

... although Downsizer Contribution still allowed after 75

The main ways to get money into super

- 1. Employer contributions**
(including Super Guarantee Contributions)
- 2. Self-employed contributions**
- 3. Salary sacrifice contributions**
- 4. Personal after-tax contributions**
(with deduction)
- 5. Personal after-tax contributions**
(with no deduction)
- 6. Government co-contributions**
- 7. Spouse contributions**
- 8. Child contributions**
- 9. Downsizer contributions**
- 10. Sale of small business**

1. Employer contributions (SGC)

Employers contribute percentage of ordinary time earnings into super

If they don't, they are required to pay a charge to the ATO, which is paid to your super
Ordinary time earnings are just your wage or salary (excluding overtime and allowances)

Minimum requirements are Superannuation Guarantee Contributions (SGC)

These are changing over the next few years and are listed on the following slide

Most employers are required to allow you to choose your own fund

Some employers are exempted when subject to State Awards

They will allocate you to a default fund if you don't make a choice

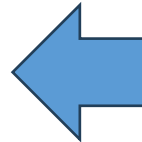
Concessional contributions subject to 15% contributions tax

Only for contributions below Concessional Contributions Cap (\$30,000 per year)

'Maximum super contributions base' for high-income earners (65,070 per quarter in 2024-25)

Super Guarantee Rates

From ...	SG Rate
1 July 2002	9.00%
1 July 2013	9.25%
1 July 2014	9.50%
1 July 2021	10.00%
1 July 2022	10.50%
1 July 2023	11.00%
1 July 2024	11.50%
1 July 2025	12.00%



2. Self-employed contributions

Self-employed people are not employees of a business

They work for themselves as a sole-trader or under a partnership

... so they don't receive Superannuation Guarantee Contributions (SGC)

... but can make their own concessional contributions themselves

1. Send a 'Notice of intent to claim' form to your super fund before end of financial year
2. Make personal contribution directly to the superannuation fund from bank account
3. Claim a tax deduction for the contribution in your Tax Return to the ATO

3. Salary sacrifice contributions

Ask your employer to pay part of your salary into super (15% contributions tax)

... instead of paying it to you (marginal income rate of tax + Medicare levy)

Employer must be notified in writing before the income is earned

Concessional contributions (before-tax)

Are not eligible for government co-contributions

Are eligible for First Home Super Saver scheme (FHSS)

Popular with people approaching retirement (aged 50 to 74)

Under-utilised by young people who could benefit from long-term compound returns

Salary sacrificing can affect termination benefits under some contracts (watch out)

Age restrictions

Not permitted for those aged 75 or above

4. Personal after-tax contributions with deduction

Payments direct from your bank account into superannuation

... for which you do claim a tax deduction in your income tax return (15% contributions tax)

Concessional Contributions subject to Cap (\$30,000 per year)

Note that Employer and Salary Sacrifice contributions are also included in this amount!

Popular way to increase super and reduce income tax

You must notify your super fund of your intent to claim a deduction and receive

NAT71121 Notice of intent to claim or vary a deduction for personal contributions

Usually eligible for First Home Super Saver (FHSS) scheme

Not eligible for Government Co-Contribution

Age restrictions

Age 67 to 74 you must satisfy 'work test' of 40 hours in consecutive 30-day period

Not permitted for those aged 75 or above

5. Personal after-tax contributions with no deduction

Payments direct from your bank account into superannuation

... for which you do not claim a tax deduction in your income tax return

Non-Concessional Contributions subject to Cap (\$120,000 per year)

\$360,000 if bring forward 2 years (balance must be below Transfer Balance Cap \$1.9 million)

Popular with people approaching retirement (aged 50 to 74)

Sell investment property and invest proceeds into superannuation

Usually eligible for First Home Super Saver (FHSS) scheme

Age restrictions

Not permitted for those aged 75 or above

6. Government super co-contributions

**A government ‘top-up’ of \$0.50 for each \$1 (up to \$500)
of personal after-tax contributions (up to \$1,000)**

Income must be below \$45,400 (2024-25) to get the full amount ... and

1. Not claim a tax deduction on the personal contribution
2. Must earn >\$1 wage income (must be >10% of total income) and lodge tax return
3. Not hold a temporary visa at any time in the financial year
4. Must be aged below 71 (no restrictions on children)
5. ... a few other things (see ATO website)

ATO and super fund arrange it automatically

Possible strategy for spouses or uni students with casual work

7. Spouse contributions

Superannuation contributions on behalf of a low-income spouse

Sum of spouse's assessable income, fringe benefits and super < \$40,000

Spouse can be married or de facto and must be living together

Both you and spouse are Australian residents

Non-concessional contribution (after-tax)

Contributor is eligible for tax rebate (maximum \$540)

Tax offset is calculated as 18% of the minimum of:

1. \$3,000 minus spouse's income over \$37,000
2. The value of the spouse contribution

... tax rebate gradually reduces to zero as spouse's income increases from \$37k to \$40k

8. Child contributions

Infants and children can have superannuation accounts

Some superannuation funds will not permit infant accounts due to inability to sign forms!

Adults can make contributions into the child or infant's accounts

Non-concessional after-tax contributions

No tax deductions can be claimed by the adults making the contributions

\$50,000 can be withdrawn for First Home Super Saver Scheme (FHSS)

The annual contribution is still limited to \$15,000 per year

... so the contributions need to be made over at least 4 years

9. Downsizer contributions

Contribute up to \$300,000 from the sale of your home into superannuation

Home must have been owned by you or spouse for 10 years prior to sale

Can only be accessed once by people aged 55 years or older

... and still can be used by those aged 75 and above

An after-tax contribution made into superannuation (no tax on main residence)

Neither a non-concessional or concessional contribution

Does not contribute to Non-concessional Contributions Cap

Can still be made even if super balance is greater than Transfer Balance Cap (\$1.9 million)

Does contribute to Transfer Balance Cap (\$1.9 million) when moving super into income phase

10. Sale of small business

Rollover proceeds from sale of small business into superannuation

... and defer tax on up to \$500,000 in capital gains

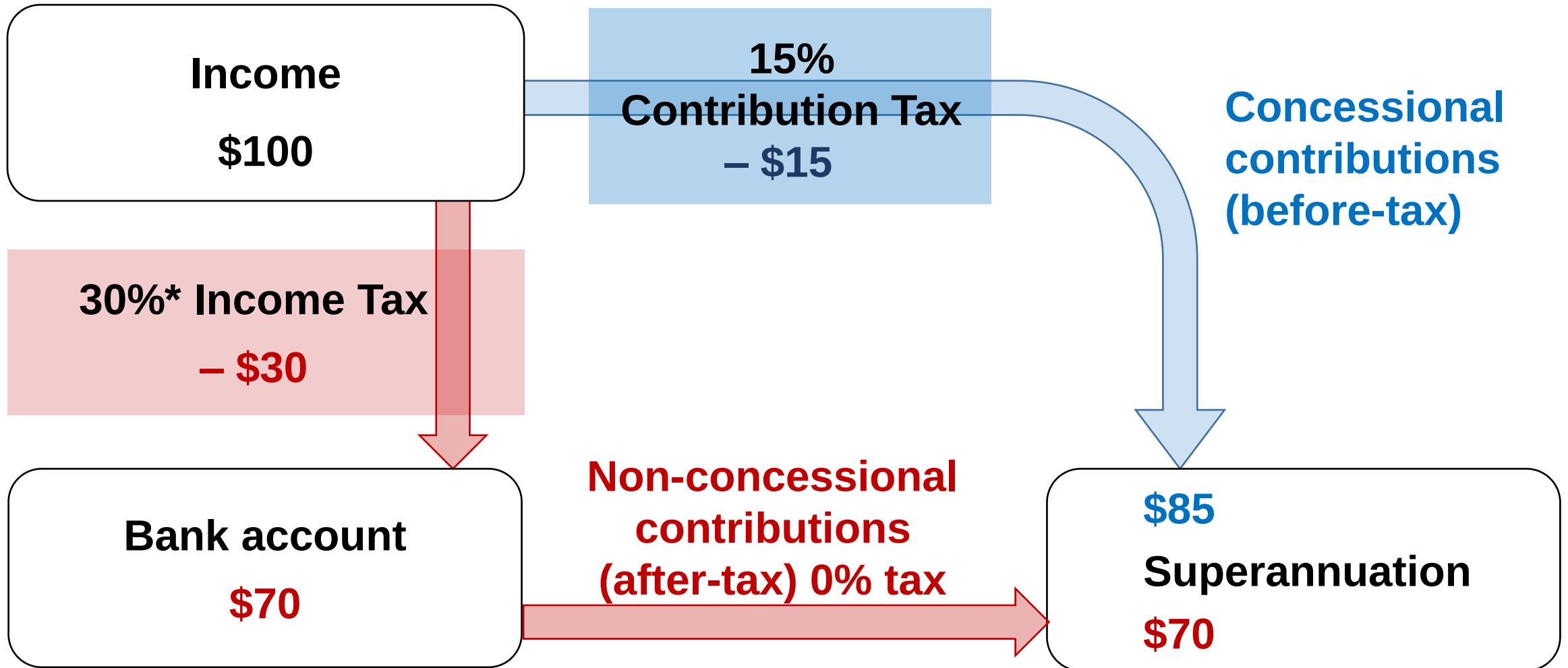
Details are complex (not covered in this course)

Summary of age restrictions for contributions

Contribution	< 55	55 - 66	67 - 74	75 +
Employer SGC (concessional contribution)	Yes	Yes	Yes	Yes
Salary sacrifice (concessional contribution)	Yes	Yes	Yes	No
Personal after-tax (with deduction) (concessional contribution)	Yes	Yes	Work Test	No
Personal after-tax (no deduction) (non-concessional contribution)	Yes	Yes	Yes	No
Downsizer (special contribution)	No	Yes	Yes	Yes

Work Test = Must work 40 hours in consecutive 30 day period

Contributions visualised



* The above Income Tax rate does not include 2% Medicare Levy (for simplicity). There is no Medicare Levy on Contribution Tax.

Drawing an Income From Super

Recap: When can I access my super?

You must reach your 'preservation age' before you can access your super

The preservation age is currently age 60

Exceptions:

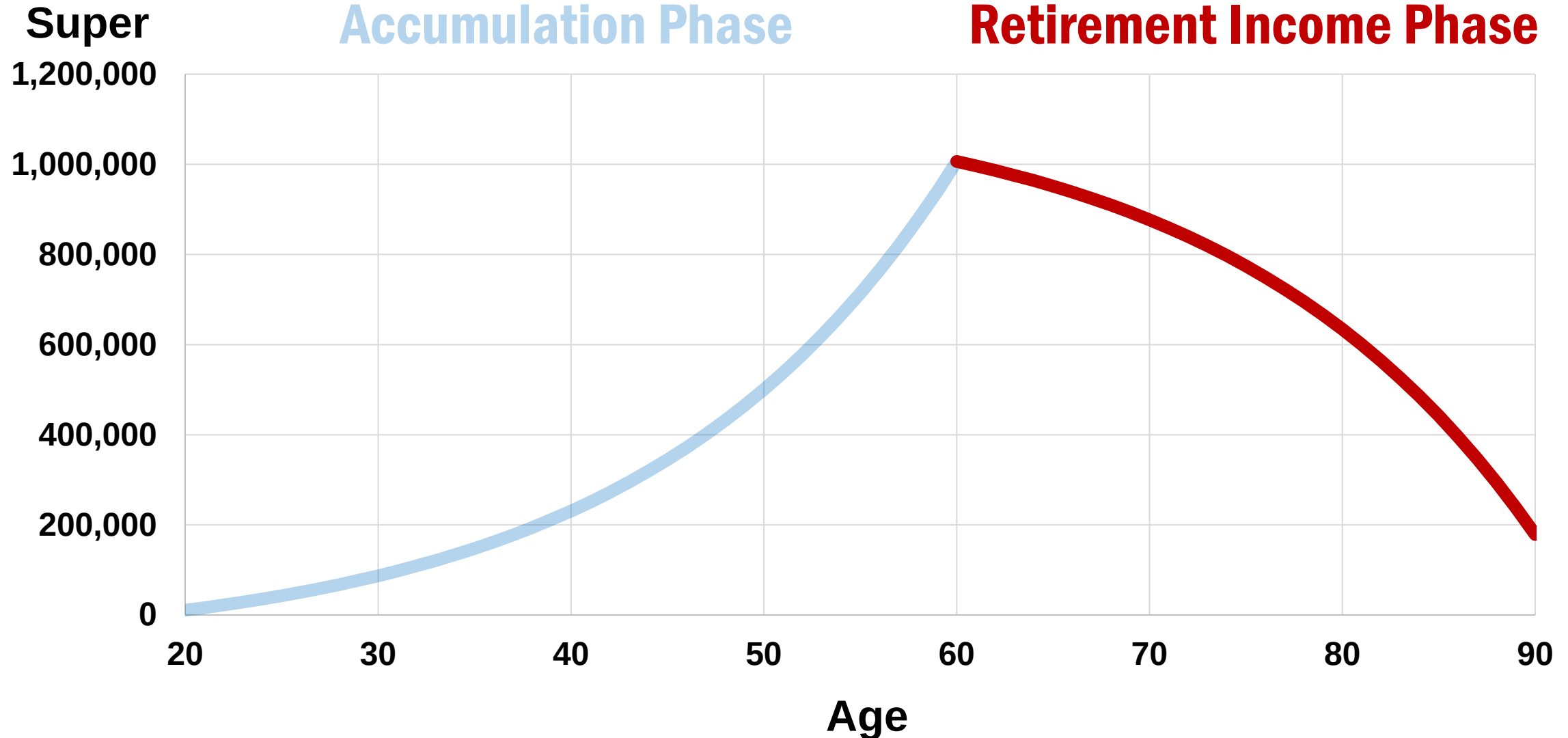
- First Home Super Saver Scheme

- Compassionate Grounds

- Financial Hardship

- and any emergency release programs (such as COVID-19 in 2020)

Focussing on retirement income phase here ...



Minimum pension payments based on age

Once a pension starts a minimum percentage of the account balance is required to be paid each financial year

Age	Percentage factors
Under 65	4%
65 to 74	5%
75 to 79	6%
80 to 84	7%
85 to 89	9%
90 to 94	11%
Aged 95 or older	14%

Transition to retirement (TTR) pension

also called a 'Transition to Retirement Income (TRI)

You can access an income stream from super while you are still working

You must reach the 'preservation age' (60)

Must draw minimum of 4% each year if under age 65

Lump-sum withdrawals not allowed unless you permanently retire

The pension income that you draw is normally tax-free

Investment earnings subject to 15% tax rate

Account Based Pensions (ABP) also called 'Allocated Pensions'

Allows you draw a regular income from super savings

... once you have reached preservation age and permanently retired

You can still make lump-sum withdrawals

If you are aged 60+ the income you receive is tax-free

Investment earnings are tax-free

Lump-sum withdrawals are usually tax-free

Maximum transferred into pension is Transfer Balance Cap (\$1.9 million)

Anything above this must be kept in super accumulation phase (earnings taxed at 15%)
... or withdrawn from super completely

Annuity

Pays a regular income regardless of market returns

Income can be linked to inflation

Term can be fixed or lifetime

Lifetime annuities can revert part of income to spouse upon death

Limited or no access to lump sum withdrawals

If you are aged 60+ the income you receive is tax-free

Investment earnings are not relevant

Maximum transferred into annuity is Transfer Balance Cap (\$1.9 million)

Anything above this must be kept in super accumulation phase (earnings taxed at 15%)
... or withdrawn from super completely

Comparison of retirement income streams

	TTR	ABP	Annuity
Preservation age	✗	✓	✓
Must be permanently retired	✗	✓	✓
Can easily make withdrawals	✗	✓	✗
Income is tax-free (age 60+)	✓	✓	✓
Investment returns tax-free	✗	✓	✓

Withdrawing Funds From Super

Why withdraw money from super?

Still have home loan payable upon retirement (unfortunate)

Don't trust the government ... get it all out! (bad reason)

Don't trust the super funds ... get it all out! (bad reason)

Health expenses

Retiring overseas (may still be better to leave it in)

Helping adult children (in 40s) reduce their debt

Helping grandchildren (in 20s) buy their first property

Lavish family holiday

Types of withdrawals

- 1. Withdrawals after age 60**
- 2. Economic crises**
- 3. First Home Super Saver scheme**
- 4. Compassionate grounds**
- 5. Severe financial hardship**
- 6. Terminal medical condition**
- 7. Incapacity**
- 8. Super less than \$200**
- 9. Departing Australia**
- 10. Divorce**
- 11. Death**

Recap: When can I access my super?

The earliest age you can access super is called your 'preservation age'

The preservation age is currently age 60.

Exceptions:

- First Home Super Saver Scheme

- Compassionate Grounds

- Financial Hardship

- Emergency Release Programs (such as COVID-19 in 2020)

What is a 'condition of release' after age 60?

Not only must you achieve your preservation age (60) ...

... you must also satisfy a 'condition of release'

- 1. Permanently retire**
- 2. Start a Transition to Retirement (TTR) pension**
- 3. Cease work with an employer**

... or reach age 65 (even if you have not retired)

1. Withdrawals after age 60

Withdrawals from super after the age of 60 are often largely tax-free

Tax-free component	0% tax
Taxable (taxed)	0% tax
Taxable (untaxed)	15% tax up to a limit then 45%

... but you need to satisfy a 'condition of release'

Such as ceasing work with an employer or permanently retiring

If you invest the funds outside the super environment

... then investment returns are assessable at marginal income tax rates

2. Economic crises

COVID-19 early release of super

\$20,000 over 2020 and 2021 financial years

Withdrawals were tax free

Provided emergency financial slack to Australians in economic crisis

Could be repeated in future economic crises

3. First Home Super Saver scheme (FHSS)

Withdraw up to \$50,000 to help buy your first home

Can only access \$15,000 of contributions per year (so must have contributed at least 4 years)

Assessed individually ... so couples can withdraw \$100,000

Must be from either:

- 1. Salary sacrifice contributions** (concessional)
- 2. Personal after-tax contributions with deduction** (concessional)
- 3. Personal after-tax contributions with no deduction** (non-concessional)

... NOT Employer contributions (concessional)

Withdrawal of concessional contributions for FHSS to buy first home

... pay marginal income tax rate less 30% tax offset when withdrawn

... save roughly 15% tax on savings channelled via super to buy first home

4. Compassionate Grounds

- 1. Medical treatment or transport for you or your dependent**
- 2. Palliative care for you or dependent**
- 3. Making payment on home loan or council rates so you don't lose home**
- 4. Accommodating a disability for you or your dependent**
- 5. Expenses associated with the death, funeral or burial of dependent**

Tax is payable on the withdrawals

Contact Australian Taxation Office

5. Severe Financial Hardship

You must meet all of the following:

1. **Unable to pay essential family living costs**
2. **Receiving income support payment for at least 26 weeks in a row**
Not including Veteran Payment, ABSTUDY, Austudy or Youth Allowance as full-time student

Tax is payable on the withdrawals

Contact your super fund

6. Terminal medical condition

1. Certified by two registered medical practitioners

One must be a specialist

2. Illness likely to result in death within 2 years

Withdrawal is tax-free

Contact your super fund

7. Temporary or permanent incapacity

Temporary incapacity

Unable to work due to physical or mental medical condition

Can generally access insurance within super

Super is received as regular payment (income stream)

Normal tax

Permanent incapacity

Permanent physical or mental medical condition

Can receive super as lump-sum or regular payment (income stream)

Concessional tax

Contact your super fund

8. Super less than \$200

Your employment is terminated and balance is below \$200

No tax is payable

Contact your super fund

9. Departing Australia super payment (DASP)

Temporary visa holders can withdraw super upon leaving Australia

DASP ordinary tax rate is 35%

Contributions tax of 15% was also paid

Total tax paid is 50%

10. Divorce

Super assets are included in the 'asset pool' distributed in a divorce

Family Court can order super funds to pay part of super to ex-spouse

Note: Relationship counselling is usually cheaper than a divorce!

11. Death

Your super is held 'in trust' by the super fund on your behalf

It is not directly owned by you so you cannot give instructions via your will

Trustee of fund is obliged to pay super to your financial dependents

If none can be found, it will be paid to your estate

You can nominate beneficiaries with your super fund

You can make 'binding nominations' but it must be to a dependent

Paid to a dependant for tax purposes: tax-free

Usually spouse (including de facto and same-sex) or child under 18 years of age

Paid to a non-dependant for tax purposes: taxed at 15% plus Medicare levy

Usually adult children or siblings

Super Strategies

1. Make additional contributions into super

Salary sacrifice at least 10% of your salary

... or make personal after-tax contributions and claim a tax deduction

Take advantage of long-term compounding of returns

Stay under the Concessional Contributions Cap (\$30,000 per year)

Note that employer contributions are included in this cap

Problems

Also need to be saving for buffer, financial slack, future expenses

No liquidity ... cannot access salary sacrificed contributions in emergency

... but you can access them to buy your first home

2. First Home Super Saver (FHSS)

Withdraw salary sacrifice or personal contributions to help buy first home

\$50,000 per individual (\$100,000 for a couple)

Only \$15,000 in contributions from any individual year can be withdrawn to buy a home

... so need to have contributed over at least 4 years

Note that concessional contributions still pay tax

Pay 15% contributions tax rather than marginal income tax rate when you make contribution

Pay marginal income tax rate less 30% tax offset when withdrawn to buy first home

... so you effectively save 15% tax on any savings channelled via super to buy first home

Problems

Employer contributions cannot be used for FHSS

Tax advantaged returns may be replicated by borrowing to invest in diversified share portfolio

Low liquidity ... can access for first home but nothing else (except for early release)

Australian Taxation Office (2024), '[First Home Super Saver scheme](#)' on [ato.gov.au](#)

3. Invest in high-growth investment option

While under 50 ... consider investing in high-growth option

Higher volatility (risk) in exchange for higher expected average return

Mainly Australian and international shares

Extra 1% per annum makes a big difference over 30+ years

4. Children's super

Open a super account for each child when they are born

Ask grandparents to make small monthly personal after-tax contribution

... preferably made via a bank account under the child's name

Take advantage of 60+ years of compounding

\$50,000 can be withdrawn for First Home Super Saver (FHSS) scheme

Contributions should be made from a bank account in child's name

5. Increase contributions once you reach 45

From age 45+ try to salary sacrifice

... or make personal after-tax contributions and claim a deduction

Watch out for the Concessional Contributions Cap (\$30,000 per year)

This reduces income tax payable

Problems

At that age you have high living expenses from private school fees, food, holidays ...

Mid-life challenges could result in career change or lavish purchases

6. Self-Managed Super Fund (SMSFs)

Create your own fund with 6 or fewer members

Before July 2021 SMSFs could only contain 4 or fewer members

Direct control over investments with more choice (including property)

Rules include 'related parties' or deriving benefits from assets (in-house assets)

Better control over tax when moving from accumulation to payment phase

Borrow to invest using limited recourse borrowing arrangement (LRBA)

Ability to transfer residual amounts to other fund members upon death

... but pretty high admin costs and an accountant needs to be involved.

7. Super balancing

Attempt to equalise account balances for you and spouse

Spouse contributions

Super splitting

An individual can split up to 85% of concessional contributions to a spouse

Receiving spouse must be under the age of 65

Non-concessional contributions cannot be split

Personal after-tax contributions

Sale of assets outside super ... direct into spouse's super account (subject to limits)

Helps to avoid breaching Transfer Balance Cap (\$1.9 million)

8. Transition to Retirement (TTR)

Many interesting strategies here!

Continue working part-time after the age of 60 as long as possible

This will allow your super balance to continue growing

Salary sacrifice income above \$45,000 into superannuation

Pay 15% contributions tax rather than 30% marginal income tax (plus Medicare levy)

Subject to concessional cap of \$30,000 per year (but can be higher with unused caps)

Draw tax-free income from TTR pension to help cover living expenses

Net effect is to lower marginal tax rate to 15% + 2% Medicare levy

9. Sell investments outside super

Before retirement ...

Simplify your investments by selling properties or shares outside super

Invest proceeds into super as personal after-tax contributions

Watch out for the Non-concessional Contributions Cap (\$120,000 per year)

... or \$360,000 in one year by bringing forward 2 years of caps

Watch out for timing of capital gains on sale of assets

Best to sell those assets in a financial year in which you are earning little other income!

10. Downsize home after age 60

After age 60, downsize the family home to a smaller one

You must have lived in the home for 10 years prior to sale

Downsizing contribution of \$300,000 into super

For a couple, this is \$600,000 in total

Watch out for Transfer Balance Cap (\$1.9 million per person)

This is still available to those aged 75 and over

11. If all else fails ... there is the Age Pension

Age Pension is a form of social security

Aged 67 or above for those born from 1 Jan 1957 onwards (from 1 July 2023 onwards)

The regular income can help your super to last a lot longer

There is both an 'asset' and 'income' test

There are limits on 'gifting' assets to others

There is 'deemed income' rates on investments

Receiving even \$1 dollar entitles you to Pensioner Concession Card

Cheaper health care, medicines and other services

12. You will not live forever in perfect health

Plan to enjoy life, hobbies, voluntary work, family and grand children

Avoid complex super arrangements, tax structures and investments

Remember that your brain will age

More confusion, more human bias, less trusting, poor memory, possible dementia

Draw a regular income

Keep the next 5 years of pension payments in low-risk (fixed interest)

Keep the remainder in growth investments (shares and listed property)

Preferably use Exchange Traded Funds to avoid gambling

Your Financial Plan

Apply what you have learned in this Unit for your financial plan:

4. Financial Independence

Describe you plan to achieve financial independence (preferably with calculations!)

Detailed description of how you plan to use super during [accumulation](#) phase

Detailed description of how you plan to use super during [payment](#) phase

More information under ‘Financial Plan Instructions’ document

... on the course website under ‘Financial Plan’ section

For assessment purposes ...

You may assume that you are living in a different country

... but you must assume the Australian superannuation system applies in that country

Join us in a Drop-in Session

Each week we have a drop-in Q&A session on Zoom

Details are on the course website under ‘Getting Started Hub’

Bring along some questions that you have about the Unit.

Sessions are not recorded.

You are also welcome to ask questions on the ‘General Forums’